August 2015

Pilot-Operated Regulator

Summary

Introduction 1
Description 1
Characteristics 2
Labelling 3
Spare Parts 4
Operation 6
Dimensions and Weights 7
Installation 8
Commissioning 9
Maintenance 10

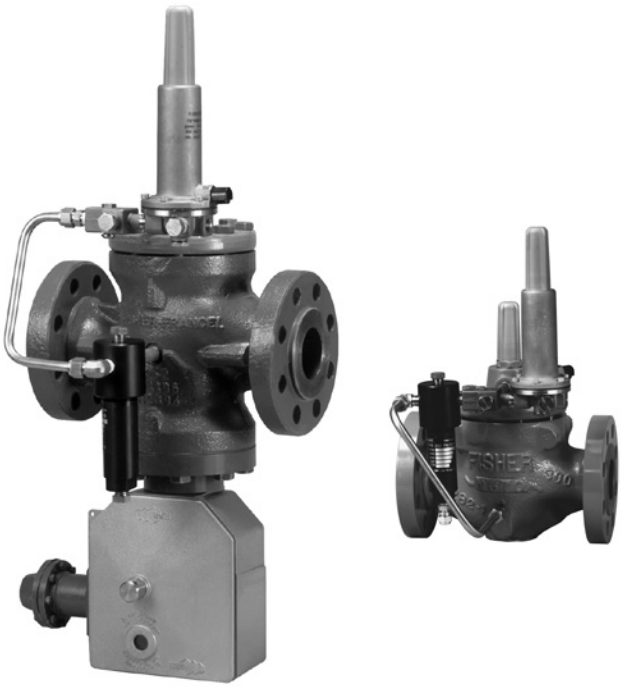


Figure 1. *Type EZR Pilot-Operated Regulator*

Europe, Middle East and Africa Document Only

Introduction

The **Type EZR** is a pilot-operated regulator used in transmission and distribution networks or pipe lines supplying industries and commercial businesses.

The **Type EZR** can be equipped with a slam shut Type OS2 (Type EZR body change) which permits the gas flow to be cut off rapidly and totally in the case of under or over outlet regulator pressure.

Description

The Type EZR consists of:

A version without integral slam shut:

- A body («E body» type), a bonnet
- A regulation subassembly consisting of a slotted cage and a diaphragm/plug
- A travel indicator, an inlet screen
- A pilot assembly consisting of a filter with bleeding screw (filtration 20 microns), an adjustable restrictor and a pilot set to the required outlet pressure

A version with integral slam shut:

- A body («X body» type), a bonnet, a connecting part
- A regulation subassembly consisting of a slotted cage and a diaphragm/plug
- A travel indicator
- A pilot assembly consisting of a filter with bleeding screw (filtration 20 microns), an adjustable restrictor and a pilot adjusted to the required outlet pressure
- A removable slam shut orifice
- A integral O-ring tightshut valve/bypass assembly
- A release relay Type OS2 according to D103683X012_ OS2_IM:
  - a mechanism box (BM)
  - a safety manometric box (BMS) to be connected outlet side of the regulator

The **Type EZR** is in conformity with the PED 97/23/EC and is classified in Category IV.

FISHER

EMERSON
Process Management

Table 1. *General Characteristics for Type EZR Regulator*

Operating pressure		PS	72.4 bar(1)	**SLAM SHUT**			
Operating temperature		TS	- 17 / 66°C(1)	**Accuracy**		AG	2.5
REGULATOR							5 (Piston)
Outlet pressure		Pa	0.5 to 48.3 bar	**Set point range**		Pt	up to 100 bar
Minimum differential		ΔP min	2 to 3 bar	**PILOT**			
Maximum operating differential	**PN 20**	ΔP max	18.6 bar	**Type of pilot**		Standard	161EB
	PN 50		50.0 bar			Monitor	161EBM
	PN 100		55.2 bar	**Manometric box**		PS BMP	52 bar
Max emergency differential		ΔP emerg	72.4 bar	**Fluid**	Groups 1 and 2 according to PED 97/23/EC 1st and 2nd family gas according to EN 437 or other gases (compressed air, nitrogen). The gas must be noncorrosive, clean (filtration on inlet side necessary) and dry.		
Accuracy		AC	2.5 - 5				

1. Values correspond to the characteristics of the regulator diaphragm.

Table 2. *Pressure and Temperature Characteristics*

Body	P_{max} (bar)	T_{min} (°C)	T_{max} (°C)
A216WCB	96.7	-20	71
A352LCC	100	-30	71

The regulator body and the slam shut have been designed to support different pressure and temperature levels

B05c

Table 3. *Flow Coefficients and Valve Plug Travel Information*

Coefficients	Capacity	DN 25	DN 50	DN 80	DN 100	DN 150
C_g	100%	480	1800	3400	5550	11200
	60%	290	1020	1970	3300	7150
	30%	140	560	970	1690	3570
C_1	100%	33	36	37	38	36
	60%	29	28	29	27	30
	30%	30	29	26	26	26
Valve plug travel (mm)		35	35	50	50	50

B05a

Table 4. *Pilot Pressure Ranges*

BMP Size	1	2	3	4	5	6	7	8
Spring Colour	White	Yellow	Black	Green	Blue	Red	Blue	Red
Spring Range (bar)	0.5	1.0	2.8	5.2	9.7	13.8	24.1	31.0
	1.0	2.8	5.2	9.7	13.8	24.1	31.0	48.3

B05b

Material

Regulator

Body	Steel
Bonnet	Steel
Slotted disc	Stainless steel
Diaphragm, O-rings	Nitrile (NBR)

Slam shut

Connecting part	Steel
Orifice	Stainless steel
Valve plug	Stainless steel

Pilot

Body	Stainless steel
Manometric box	Stainless steel or Aluminium
Restrictor	Stainless steel
Filter	Aluminium
Cartridge	Polythene

Connections

Inlet / Outlet: ISO PN 100 B (ANSI 600 RF)
ISO PN 50 B (ANSI 300 RF)
ISO PN 20 B (ANSI 150 RF)
Other possibilities exist (contact factory)
ISO PN 16 B, 25 B, 40 B

Pilot Impulse line (IP): 1/4'' NPT tapped
Pilot Monitor impulse line (IM): 1/4'' NPT tapped
Intermediate impulse line (PI): 1/4'' NPT tapped
Slam shut impulse line (IS): 1/4'' NPT tapped
Mechanism box vent (E): 1/4'' NPT tapped
Impulse diameter: Pipe interior Ø 8/10mm min.

Europe, Middle East and Africa Document Only

Labelling

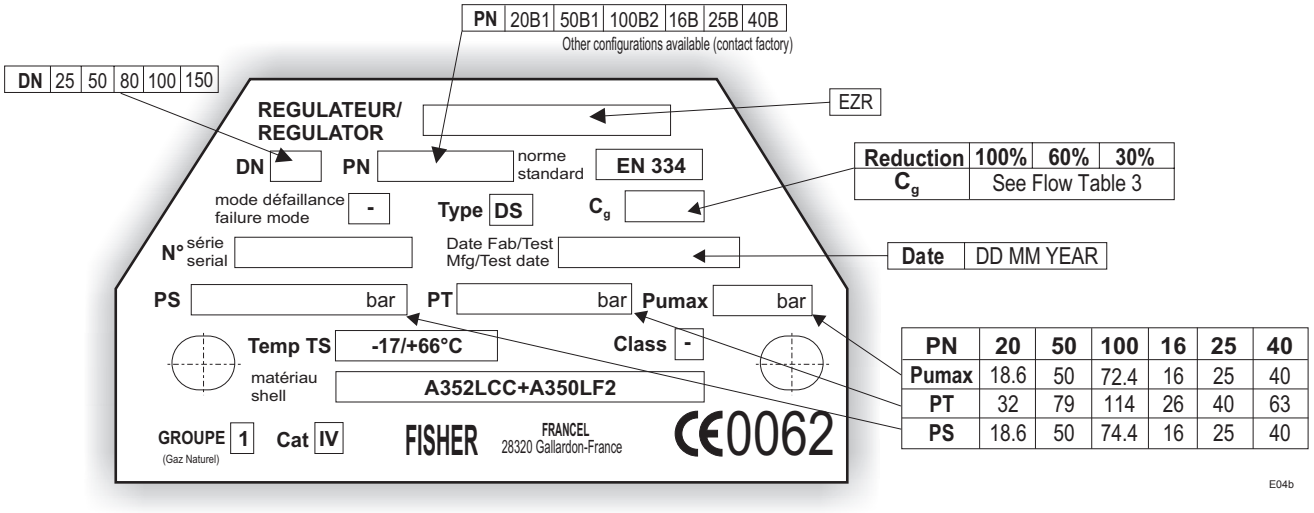


Figure 2. *Type EZR Regulator Label*

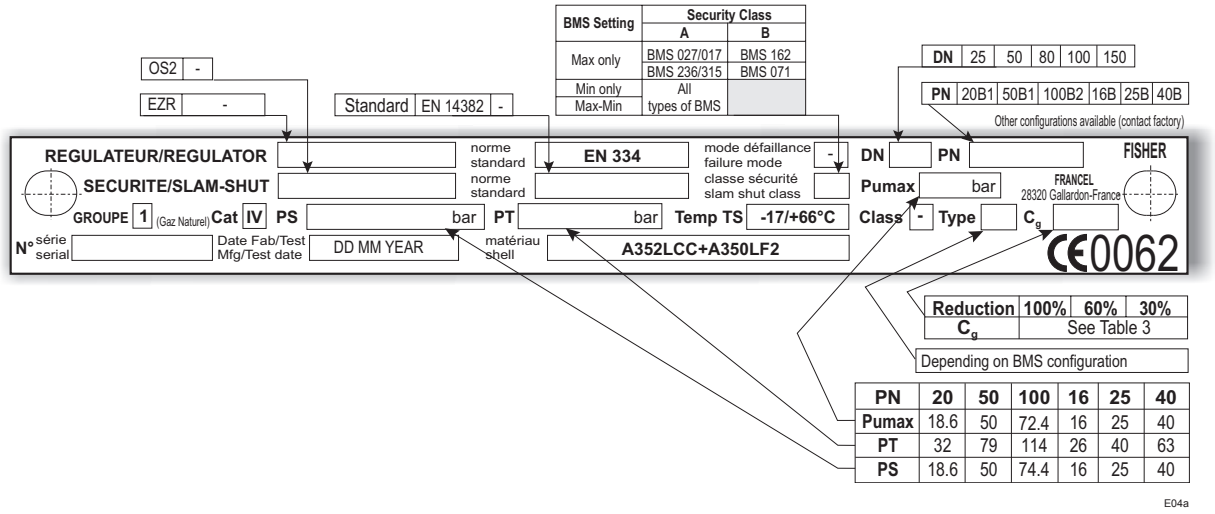


Figure 3. *Type EZR Regulator with Type OS2 Slam Shut Label*

Europe, Middle East and Africa Document Only

Spare Parts

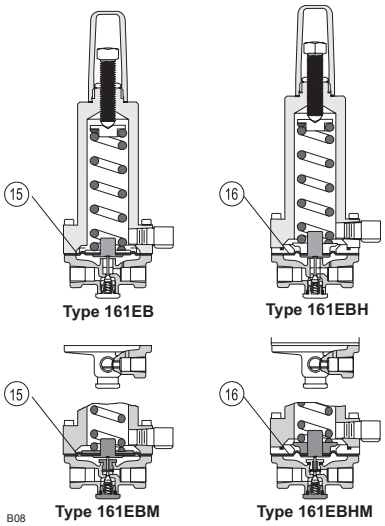


Figure 4. *Types 161EB, 161EBH, 161EBM and 161EBHM Pilots*

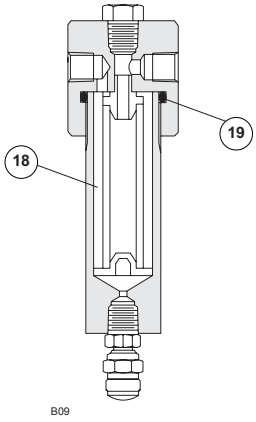


Figure 5. *Filter*

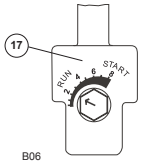


Figure 6. *Adjustable Restrictor*

Table 5. *Pilot, Restrictor and Filter Spare Parts List*

B11

Description	Key	Reference			
		Pilot Type			
		161EB	161EBH	161EBM	161EBHM
Pilot kit 0.34 to 13.8 bar	15	R161X000012	----------	R161MX00012	----------
Pilot kit 13.8 to 24.1 bar	15	R161X000022	----------	R161MX00022	----------
Pilot kit EBH(M)	16	----------	R161HX00012	----------	R161HMX0012
O-ring	17	----------	1C8538X0052		----------
Filter cartridge	18	----------	17B6813X012		----------
O-ring	19	----------	1F269206992		----------
SAV kit 0.34 to 13.8 bar(1)	----	197435	----------	197438	----------
SAV kit 13.8 to 24.1 bar(1)	----	197436	----------	197439	----------
SAV kit 24.1 to 48.3 bar(1)	----	----------	197437	----------	197440

1. The SAV kits include item numbers 15 or 16, 17, 18 and 19.

BMP Size	Spring Reference
1	17B1260X012
2	17B1262X012
3	17B1259X012
4	17B1261X012
5	17B1263X012
6	17B1264X012
7	17B1263X012
8	17B1264X012

To pass from 0.5 to 13.8 bar or from 24.1 to 48.3 bar the spring must be changed.

Europe, Middle East and Africa Document Only

Spare Parts

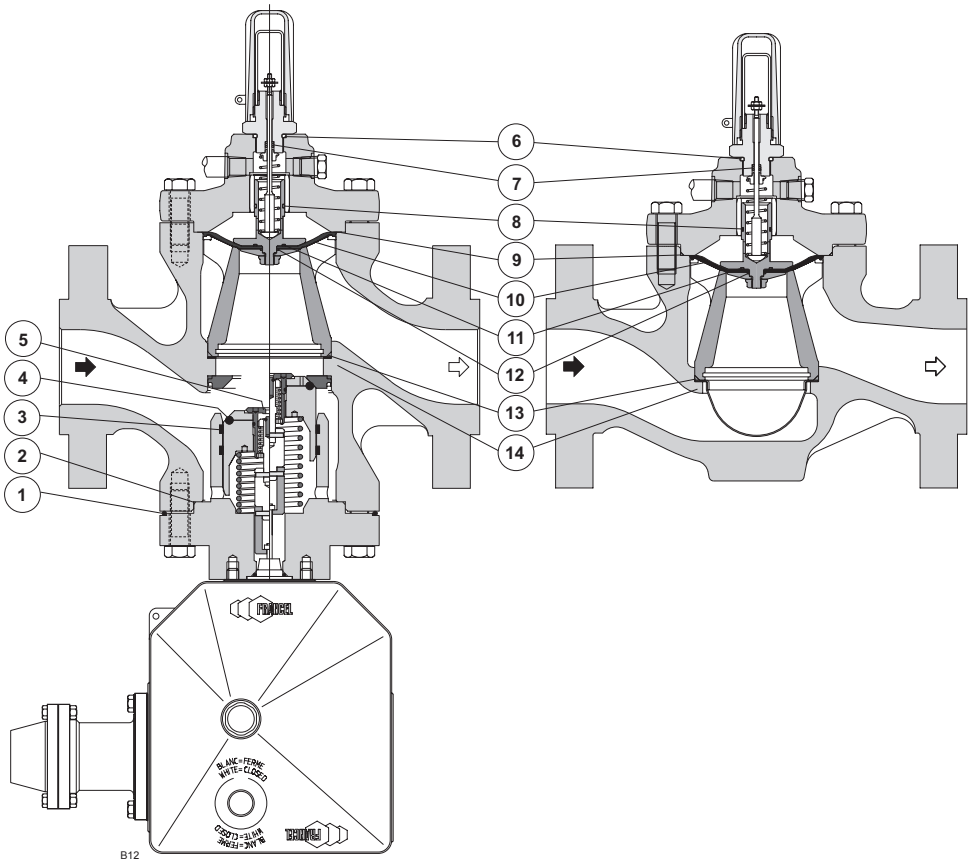


Figure 7. *Types EZR and EZR-OS2 Spare Parts*

Table 6. *Types EZR and EZR-OS2 Spare Parts List*

Regulator, Type EZR	Description	Key	Reference				
			DN / inches				
			25 / 1"	50 / 2"	80 / 3"	100 / 4"	150 / 6"
with slam shut	O-ring	1	400009	400024	400091	400045	400262
	O-ring	2	19B2838X012	18B2124X012	18B8514X012	18B2140X012	19B0359X012
	Guide	3	401950	401951	401952	401953	401954
	O-ring	4	400527	400263	400258	400260	400261
	Bypass	5	180977	180977	180977	180977	180977
	SAV kit PN 20	1 to 14	197421	197422	197424	197425	197427
	SAV kit PN 50/100	1 to 14	197421	197423	197424	197426	197427
without slam shut	SAV kit PN 20	6 to 14	197428	197429	197431	197432	197434
	SAV kit PN 50/100	6 to 14	197428	197430	197431	197433	197434
with/without slam shut	O-ring	6	18B3438X012	18B3438X012	10A8931X012	10A8931X012	10A3800X012
	O-ring	7	1H2926X0032	1H2926X0032	1D191706992	1D191706992	1D191706992
	O-ring	8	13A1584X052	13A1584X052	10A3803X062	10A3803X062	T12050X0012
	O-ring	9	19B2838X012	18B2124X012	18B8514X012	18B2140X012	19B0359X012
	PN 20 diaphragm	10	39B2397X012	29B2715X022	39B2726X012	38B5965X012	49B0357X012
	PN 50/100 diaphragm	10	39B2397X012	28B2123X052	39B2726X012	39B3996X012	49B0357X012
	O-ring	11	13A1584X052	13A1584X052	10A3803X062	10A3803X062	T12050X0012
	O-ring	12	1E216306992	1E216306992	1J4888X0052	1J4888X0052	11A8741X052
	O-ring	13	14A5713X012	10B4428X012	10B4366X012	10B4373X012	1H862306992
	O-ring	14	- - - - - - - - - - -	- - - - - - - - - - -	- - - - - - - - - - -	- - - - - - - - - - -	1D269206992
	Safety manometric box	- - - -	See D103683X012_OS2_IM manual				

B13

Europe, Middle East and Africa Document Only

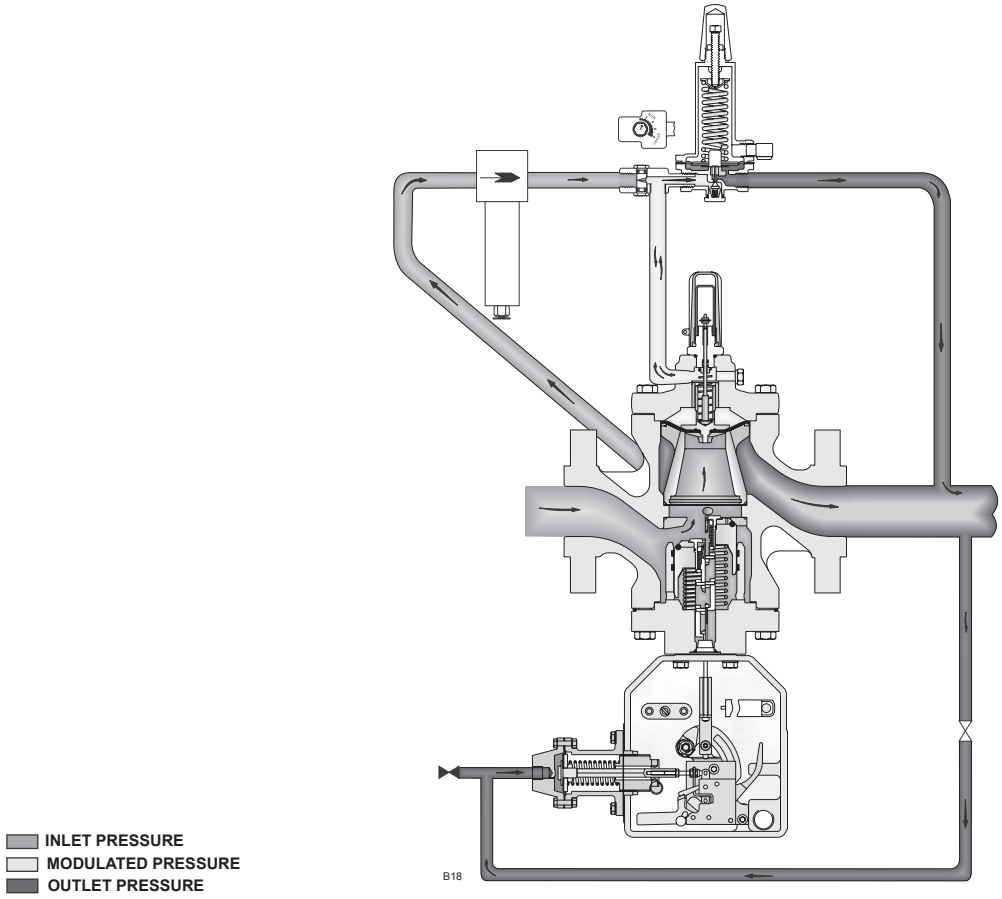


Figure 8. *Type EZR Operational Schematic*

Operation

Regulator

The Type EZR is a pilot-driven, diaphragm/plug regulator.

Tight shutoff is achieved by the diaphragm/plug pushing against the slotted cage, the force of the closing spring and the inlet pressure.

- **Opening**

  As the flow increases, the outlet pressure Pd decreases on the outlet side of the regulator and on the pilot diaphragm.

  Due to the force of the spring, the pilot opens.

  The pilot flow increases, the pressure loss through the pilot restrictor increases.

  The modulated pressure Pm decreases.

  The force of the closing spring and that of the Pm becomes inferior to that provoked by the Pu, the regulator OPENS.

- **Closing**

  As the flow decreases, the Pd increases outlet side of the regulator.

  The force of the pilot diaphragm is overcome by the force of the spring, the pilot closes.

  The pressure loss through the pilot restrictor decreases.

  The force of the closing spring and that of the Pm becomes superior to that provoked by the Pu, the regulator CLOSES.

Slam Shut

The pressure of the zone to be protected (generally the pipeline on the outlet side of the regulator and after the slam shut) is sensed by the safety manometric box (BMS).

If the pressure exceeds the set tripping pressure, the release relay frees the valve plug.

Due to the force of the closing spring and the fluid (trying to close), the valve plug closes on the orifice.

The gas flow is obstructed until the fault has been corrected and the mechanism box manually rearmed.

To reopen the valve plug an equal pressure balance on inlet and outlet sides of the regulator is required.

The mechanism box is rearmed after opening the internal bypass.

Rearming and balancing are achieved at the same time.

Europe, Middle East and Africa Document Only

Dimensions and Weights

Table 7. *Dimensions (mm) - Type EZR-OS2 (part 1)*

Body DN	B	C	E	H
25	233	315	348	250
50	243	330	357	265
80	361	366	410	301
100	393	410	454	345
150	423	396	468	332

B15b

Table 8. *Weights (kg) - Type EZR-OS2*

Body DN	PN 20	PN 50	PN 100
25	20	21	22
50	39	41	43
80	63	69	71
100	104	113	123
150	192	211	244

B16b

Table 9. *Dimensions (mm) - Type EZR (part 1)*

Body DN	B	C	E
25	220	62	335
50	226	83	340
80	343	105	392
100	372	137	433
150	420	178	465

B15a

Table 10. *Weights (kg) - Type EZR*

Body DN	PN 20	PN 50	PN 100
25	12	14	16
50	26	27	31
80	50	51	57
100	67	73	88
150	97	108	161

B16a

Table 11. *Dimensions (mm) - Types EZR and EZR-OS2 (part 2)*

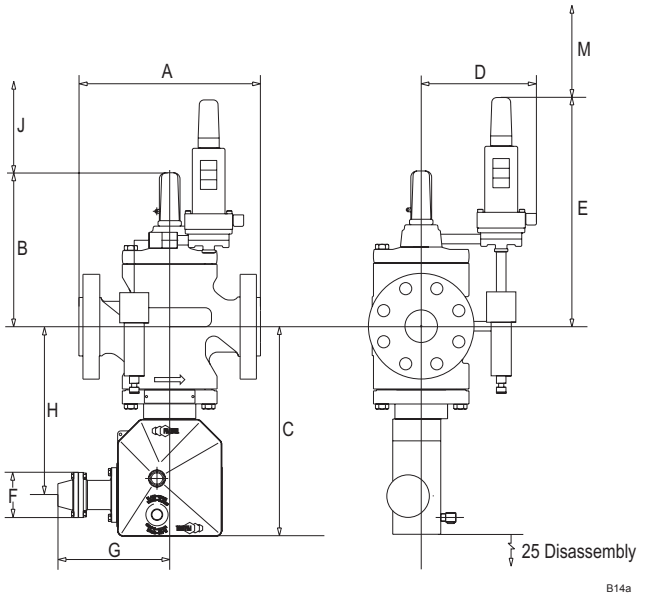
Body DN	A						D	J	M
	PN								
	16	20	25	40	50	100			
25	193.5	184	193.5	193.5	197	210	165	68	54
50	254	254	267	267	267	286	165	68	54
80	310	298	317	317	317	337	181	95	54
100	350	352	368	368	368	394	187	95	54
150	451	451	473	473	473	508	249	95	54

B17a

Table 12. *Dimensions (mm) - BMS (Safety Manometric Box)*

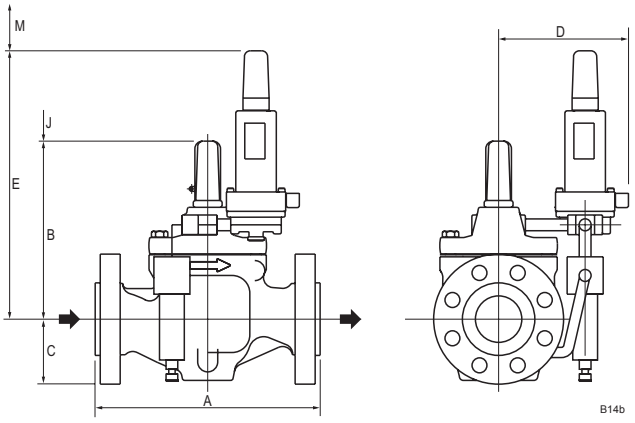
Type	F	G
Diaphragm	162	181
Piston	71	204
Bellows	74	223

B17b



B14a

Figure 9. *Dimensions (mm) - Type EZR-OS2*



B14b

Figure 10. *Dimensions (mm) - Type EZR*

Europe, Middle East and Africa Document Only

Installation

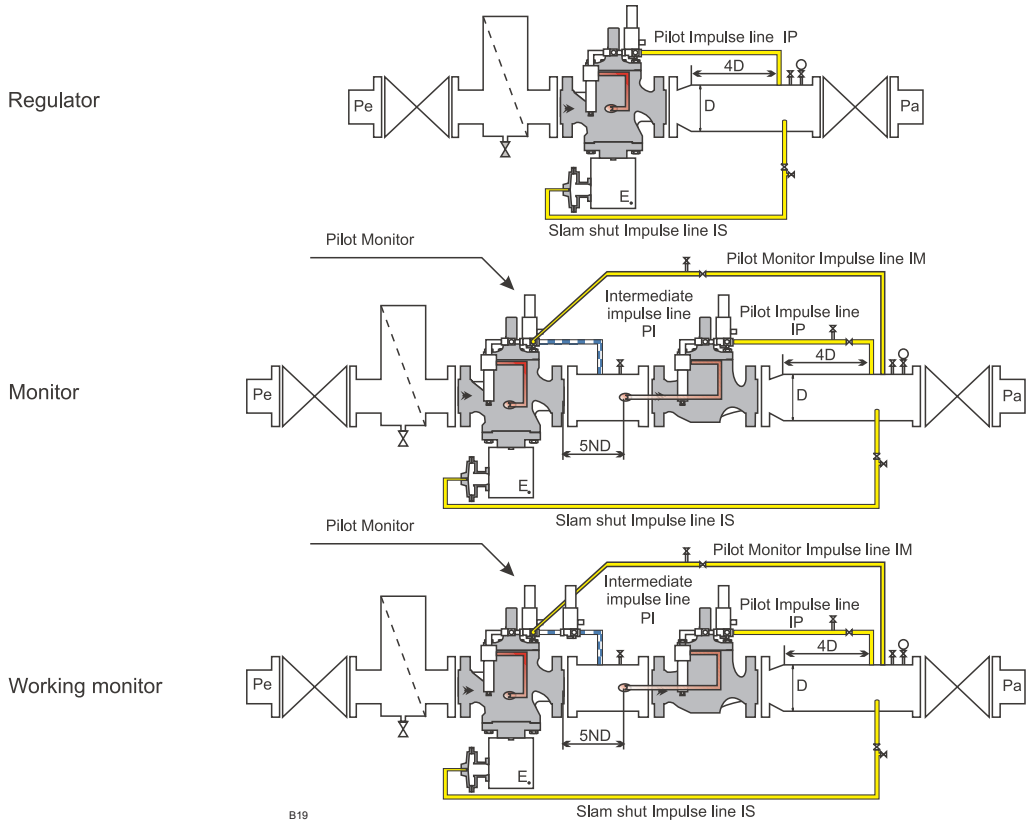


Figure 11. *Type EZH Installation Configurations*

All interventions on the equipment should only be performed by qualified and trained personnel.

WARNING

- **The regulator is installed on horizontal pipeline. Version with slam shut, the release relay is situated towards the bottom (see schematic).**
- **Installation according to EN12186 recommended.**
- **Install according to direction of fluid flow (arrow).**
- **When assembling with adjacent elements care must be taken not to create pressure force on the body and the assembling elements (bolts, O-rings, flanges) should be compatible with the geometry and working conditions of the equipment.**
- **If the case arises a support must be used to avoid pressure force on the body (a support can be installed under the flanges).**
- **Version with integral slam shut , connect the safety manometric box (IS) to the impulse at 4D on a straight run of the outlet pipe.**
- **It is recommended to separate the slam shut impulse line (IS) from that of the pilot (IP). Do not connect the impulses on the lower generator line.**
- **It is recommended to install an isolation valve and an atmospheric valve, which can be useful for slam shut tripping and verifications.**
- **No modification should be made to the structure of the equipment (drilling, grinding, soldering...).**

WARNING

- **It is recommended to install a servicing valve on the outlet pipeline to facilitate adjustments and bleeding off to the atmosphere.**
- **Verify that the inlet side is protected by an appropriate device(s) to avoid exceeding the limits of utilization (PS, TS).**
- **Verify that the limits of utilization correspond to the appropriate operating conditions.**
- **Version with integral slam shut, verify that the safety manometric box (BMS) and spring correspond to the appropriate operating conditions on the outlet side of the regulator.**
- **The equipment should not receive any type of shock, especially the release relay.**
- **The user should verify or carry out a protection adapted to the environment.**
- **Fire, seismic and lightening are not taken into consideration in standard regulators. If required, a special product selection and/or specific calculations may be supplied according to specific requirements.**
- **Version without integral slam shut, verify that a pressure limiting device on the outlet side of the regulator guarantees a pressure limit inferior or equal to the pilot PS.**

Europe, Middle East and Africa Document Only

Commissioning

Operations concerning the integral slam shut version are in italic.

All interventions on equipment should only be performed by qualified personnel

Preliminary Verifications

Start-up positions

- Inlet and outlet valves
  → Closed
  Verify the absence of pressure between inlet and outlet valves
- *Slam shut valve plug*
  → **Closed**
- Pilot **A**
  → Unloaded
- Restriction **B**
  → START position

Slam shut set point verification

Using the atmospheric valve, inject a pressure equal to the pressure required for the regulator

- *1st release relay stage*
  → Set (Stage 1)
- *Slam shut valve plug*
  → *Open (Stages 2 and 3)*
  → *Progressively increase the pressure to reach tripping*
  → *Adjust setting if necessary (D103683X012_OS2_IM)*
  Note the set point value on the equipment
  or mark it on a commissioning document

Positions before commissioning

- *Impulse line isolating valve*
  → *Open*
- *Impulse line atmospheric valve*
  → ***Closed***
- *Slam shut valve plug*
  → ***Closed***
- Servicing valve
  → Closed

The equipment is ready for commissioning

Commissioning (Act slowly)

- Inlet valve
  → Open **very** slowly
- *1st release relay stage*
  → *Set (Stage 1)*
- *Slam shut valve plug*
  → *Bypass (Stage 2)*
  → *Open (Stage 3)*

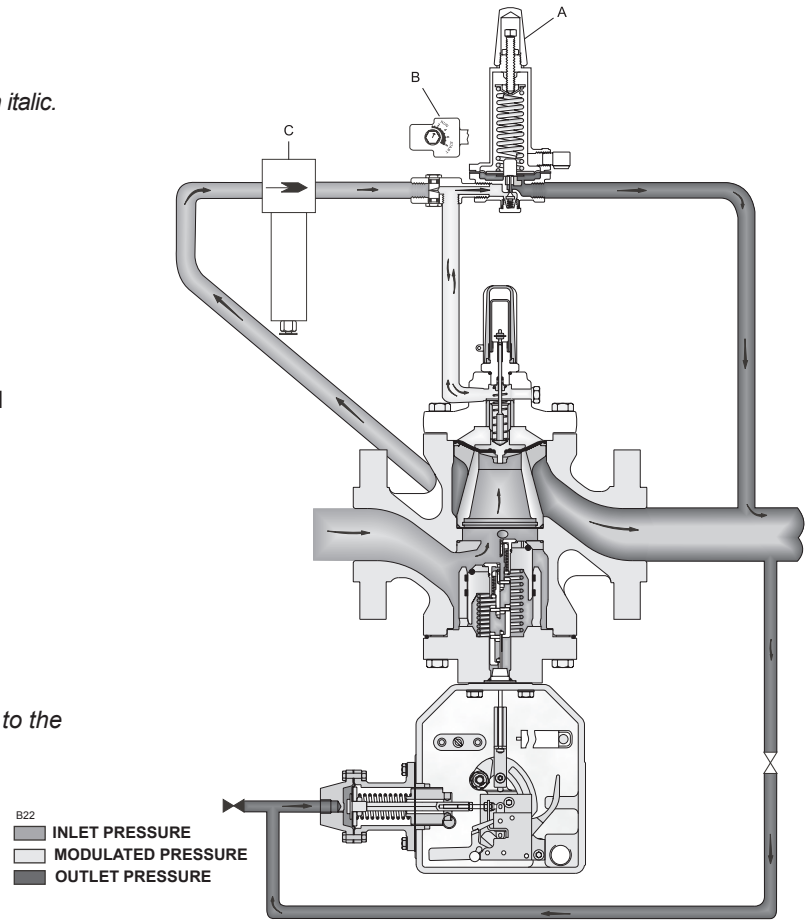


Figure 12. *Type EZH-OS2 Regulator*

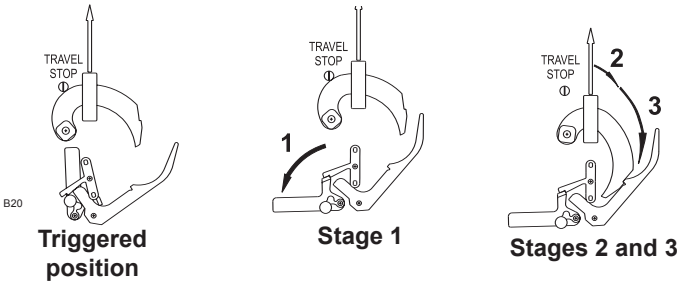


Figure 13. *Setpoint Verification Phase*

- Servicing valve
  → Open slightly
- Pilot
  → Screw to set outlet pressure
- Outlet valve
  → Open slowly
- Restriction
  → Set to "RUN" by successive fractions 2, 4 or 6
- Servicing valve
  → Closed

The equipment is commissioned

It is recommended to seal the release relay

Europe, Middle East and Africa Document Only

Maintenance

Operations concerning the integral slam shut version are in italic.

Servicing Check

Recommended frequency:

- Twice yearly minimum

Verification:

- Set point verification
- Regulator valve plug tightness
- *Tripping and set point value*
- *Slam shut valve tightness*

Departure positions

- Inlet valve → Open
- Outlet valve → Open
- *Slam shut valve plug → Open*
- Regulator → In operation

Inlet and outlet sides of regulator under pressure

Tightshut verification *(and tripping verification for integral slam shut versions)*

- Inlet valve → Closed
- Outlet valve → Closed
- Regulator — Observe the evolution of the outlet pressure (control regulator tightness)

Table 12. *Troubleshooting for Types EZR and EZR-OS2 Regulators*

If the outlet pressure increases	**Internal leak** Control the regulator valve plug Control the regulator orifice Control the pilot	**or contact after-sales**
If the outlet pressure decreases	**External leak** Locate and seal the leak	**or contact after-sales**
If the outlet pressure is constant	***The regulator is tightshut*** *Increase the set point until tripping occurs (without exceeding the outlet limits)*	
If the slam shut valve plug will not close	***Operating fault*** *Control the release relay* *Control the slam shut valve plug*	***or contact after-sales***
If the slam shut valve plug closes *Observe the evolution of the outlet pressure (control tightness)*	***Operating correctly***	
If the outlet pressure is constant Observe the evolution of the outlet pressure (control tightness)	Purge the outlet side of the regulator	
If the outlet pressure increases	***Internal leak*** *Control the slam shut valve plug* *Control the slam shut orifice* *Control the bypass*	***or contact after-sales***
If the outlet pressure is constant	***Slam shut valve plug is tightshut***	

Filter verification

- Purge the filter **C**
- Verify the cartridge

Europe, Middle East and Africa Document Only

Maintenance (continued)

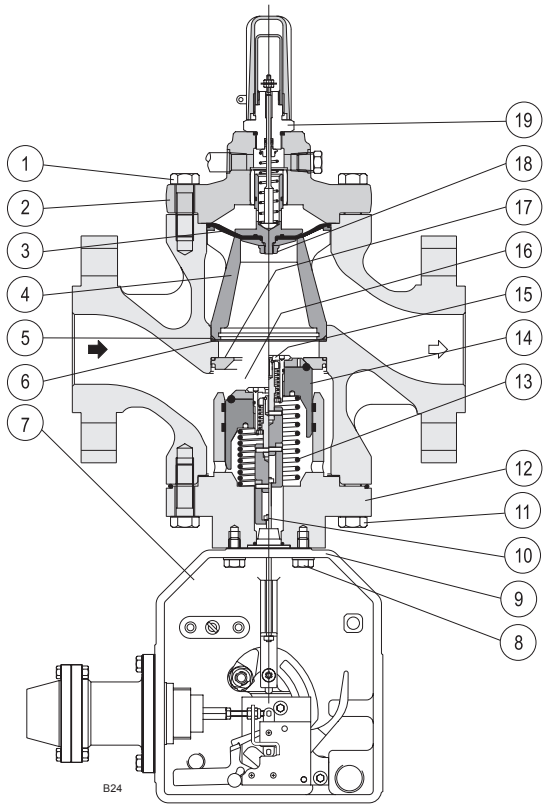


Figure 14. *Type EZR-OS2*

Table 13. *Torque Specifications*

Body DN / Inches	Screw keys 1 and 11	Spanner (inches)	Screw keys 1 and 11	Torque N.m.		
				Fixation keys 18	Connector key 19	Bypass key 15
25 / 1"	9/16 - 12 x 1 3/4	13/16	110	8	130	14
50 / 2"	1/2 - 13 x 1 1/2	3/4	110	9	130	14
80 / 3"	5/8 - 11 x 1 3/4	15/16	175	28	280	20
100 / 4"	3/4 - 10 x 2 1/4	1 1/8	260	28	280	24
150 / 6"	1 - 8 x 2 3/4	1 1/2	510	70	410	24

B25a B25b

Disassembly

Recommended frequency:
Every 2 to 6 years (or less depending on operating conditions)

Verification:
Condition of O-rings, diaphragms, lubrication

Replacement:
O-rings, diaphragm

Tools:
Dimensions according to tables below

Preliminary Operations

- Valve plug closed
- Inlet and outlet valves closed
- **Bleed off outlet pressure**
- **Bleed off inlet pressure**
- Unscrew the pilot impulse connection
- Unscrew the screws **(key 1)** fixing the bonnet **(key 2)**
- Remove the bonnet **(key 2)**
- Remove the diaphragm/plug assembly **(key 3)**
- Remove the slotted cage **(key 4)**, the O-ring **(key 5)**, the strainer **(key 6)** *(or the space washer **(key 6)**)*
- Clean parts and replace them if necessary

Pilot

- Unscrew the manometric box screws
- Remove the diaphragm

Slam Shut (Version with slam shut)

- Unscrew the BMS impulse line connector (IS)
- Remove the BM cover **(key 7)**
- Unscrew the BM fixing screw **(key 8)**
- Remove the holding pin **(key 10)**
- Remove the BM **(key 9)**
- Unscrew the screws **(key 11)** from the connecting part **(key 12)**
- Remove the connecting part **(key 12)**
- Remove the spring **(key 13)** and the valve plug **(key 14)**
- Unscrew the bypass **(key 15)**
- Unscrew the screws CHC **(key 16)** (DN 100 and 150)

Removing the orifice **(key 17)** (not recommended) requires a special extraction tool

Europe, Middle East and Africa Document Only

Maintenance (continued)

Reassembly

- Perform the above operations in reverse order (respect tightening torques)
- Replace the O-rings and diaphragm at each disassembly

Slam Shut Reassembly (Version with slam shut)

- The valve plug should be held in an upper position using a packing gland and a box to facilitate reassembly
- Precaution must be taken concerning the passage of the valve plug over the segments
- Lubricate screws before tightening
- Lightly lubricate the O-rings (silicone grease) except for the valve plug O-ring
- Lightly lubricate the stem (silicone grease)
- Lubricate the release relay mechanism (yoke and bolt) (molybdenum graphite grease)
- Lubricate the BMS spring (molybdenum graphite grease)
- A special tool is required for reassembling a new orifice

Industrial Regulators

Emerson Process Management
Regulator Technologies, Inc.

USA - Headquarters
McKinney, Texas 75070, USA
Tel: +1 800 558 5853
Outside U.S. +1 972 548 3574

Asia-Pacific
Shanghai 201206, China
Tel: +86 21 2892 9000

Europe
Bologna 40013, Italy
Tel: +39 051 419 0611

Middle East and Africa
Dubai, United Arab Emirates
Tel: +971 4811 8100

Natural Gas Technologies

Emerson Process Management
Regulator Technologies, Inc.

USA - Headquarters
McKinney, Texas 75070, USA
Tel: +1 800 558 5853
Outside U.S. +1 972 548 3574

Asia-Pacific
Singapore 128461, Singapore
Tel: +65 6777 8337

Europe
O.M.T. Tartarini s.r.l. Via P. Fabbri 1,
I-40013 Castel Maggiore (Bologna), Italy
Tel: +39 051 419 0611
Francel SAS, 3 ave Victor Hugo,
CS 80125 - Chartres 28008, France
Tel: +33 2 37 33 47 00

Middle East and Africa
Dubai, United Arab Emirates
Tel: +971 4811 8100

TESCOM

Emerson Process Management
Tescom Corporation

USA - Headquarters
Elk River, Minnesota 55330-2445, USA
Tels: +1 763 241 3238
+1 800 447 1250

Asia-Pacific
Shangai 201206, China
Tel: +86 21 2892 9499

Europe
Selmsdorf 23923, Germany
Tel: +49 38823 31 287

Europe, Middle East and Africa Document Only

For further information visit www.emersonprocess.com/regulators

The Emerson logo is a trademark and service mark of Emerson Electric Co. All other marks are the property of their prospective owners. Fisher is a mark owned by Fisher Controls, Inc., a business of Emerson Process Management.

The contents of this publication are presented for informational purposes only, and while every effort has been made to ensure their accuracy, they are not to be construed as warranties or guarantees, express or implied, regarding the products or services described herein or their use or applicability. We reserve the right to modify or improve the designs or specifications of such products at any time without notice.

Emerson Process Management does not assume responsibility for the selection, use or maintenance of any product. Responsibility for proper selection, use and maintenance of any Emerson Process Management product remains solely with the purchaser.

***O.M.T. Officina Meccanica Tartarini S.R.L.**, R.E.A 184221 BO Cod. Fisc. 00623720372 Part. IVA 00519501209 N° IVA CEE IT 00519501209, Cap. Soc. 1.548 000 Euro i.v. R.I. 00623720372 - M BO 020330*

***Francel SAS**, SIRET 552 068 637 00057 APE 2651B, N° TVA : FR84552068637, RCS Chartres B 552 068 637, SAS capital 534 400 Euro*

©Emerson Process Management Regulator Technologies, Inc., 2009, 2015; All Rights Reserved

EMERSON
Process Management